



CHINGAARI · AI TRACK · IBM BOB HACKATHON

AURA Grid

Predictive failure intelligence for transformers — from raw telemetry to a crew standing at the right door, before the outage does.

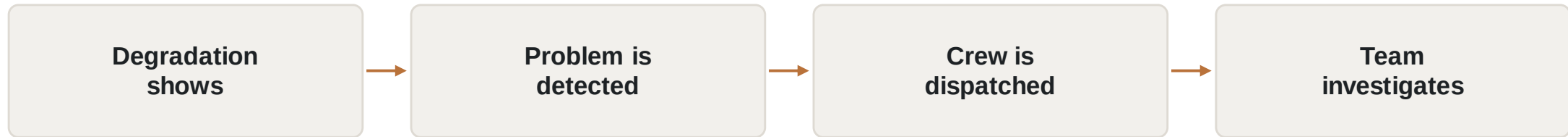
WHAT IT DOES

7-day failure forecasting · multi-factor risk scoring · maintenance advisor
crew deployment optimizer · grounded AI assistant (IBM Bob)

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By the time the alarm rings, the outage has already started

Utility teams already have what they need — sensor readings, transformer data, weather, maintenance history, incidents and crew rosters. It just lives in different systems, so the response stays reactive:



TODAY'S CYCLE — ALWAYS ONE STEP BEHIND

◆ *“What if the same data could show which transformer is likely to fail before the failure happens?”*

— the question AURA Grid is built to answer

Four teams, one blind spot

◆ Utility Operators

Which transformers are at risk right now, and which failure would hurt most?

◆ Reliability Engineers

Condition-based planning that explains why an asset is getting risky.

◆ Crew Dispatch Managers

Who is qualified, available, and close enough to respond fast?

◆ Decision Makers

Grid-wide risk, critical assets and resourcing, at a glance.

WHY IT'S HARD — THE SAME SIGNAL, SIX DIFFERENT SYSTEMS

Sensor Telemetry

Transformer Metadata

Weather Data

Maintenance History

Incident Log

Crew Roster

An alarm answers “what is happening now” — not “is this likely to fail soon.”

One pipeline: raw signal to grounded decision

AURA Grid fuses transformer telemetry, asset context, weather, history and crew data into a single decision-support workflow — from a raw reading to a ranked, actionable recommendation.



“AURA Grid connects what might happen with what should we do about it.”

TWO WAYS IN

Individual Asset

Enter a transformer ID → full prediction, risk, advisor and crew match for that asset.

Batch Analysis

Upload a CSV (up to 500 rows) → every transformer scored and ranked in one pass.

What's actually built

01

7-Day Failure Forecast

XGBoost model over fused sensor, weather and history features — a failure probability from 0.0 to 1.0.

02

Multi-Factor Risk Score

Blends failure probability with grid impact, asset criticality and weather risk — not probability alone.

03

Maintenance Advisor

Rule-based engine turns abnormal readings into a concrete, explainable recommendation.

04

Crew Optimizer

Ranks available crews by certification, skill, location and response history.

05

Live + Batch Input

Manual telemetry entry for one asset, or CSV upload for fleet-wide analysis.

06

AI Assistant

Natural-language operator Q&A, grounded in live prediction and crew data.

Eight layers, one responsibility each

01	Input	Sensor, metadata, weather, maintenance, incident & crew data	CSV / JSON
02	Processing	Validation, type conversion, feature engineering (Pandas)	Pandas
03	ML & Decision	Failure probability + risk scoring, with heuristic fallback	XGBoost
04	Recommendation	Maintenance advisor rules + crew optimizer ranking	Rule Engine
05	Database	Structured operational data & prediction history	PostgreSQL 18
06	Backend API	10 endpoints — predict, batch, risk-assets, crew, Bob	FastAPI
07	Frontend	Risk dashboard, telemetry input, batch upload, AI chat	React + Vite
08	AI Assistant	Operator Q&A grounded in current operational state	Gemini AI

A score that reflects consequence, not just probability

RISK SCORE FORMULA

0.40 * Failure Probability
+
0.30 * Grid Impact
+
0.20 * Asset Criticality
+
0.10 * Weather Risk

RISK ZONES



0.72 — TR034

LOW

< 0.30

Routine maintenance

MEDIUM

0.30 — 0.60

Monitor, plan within 1 week

HIGH

0.60 — 0.80

Schedule within 24 hours

CRITICAL

≥ 0.80

Immediate dispatch

Two transformers can share a failure probability and still deserve very different priority — grid impact and criticality decide which one gets the crew first.

IMPLEMENTATION NOTE

Weights and thresholds are configuration, not fixed law — they are tuned and kept in sync with the prediction code as real outage data becomes available, not hard-coded assumptions.

Two decisions, cleanly separated

Machine learning estimates risk. Rule-based logic decides what to do about it — kept apart on purpose, so each stays explainable.

MAINTENANCE ADVISOR · RULE-BASED

◆ Temp > 90°C AND vibration > 5 mm/s

→ Thermal + mechanical stress inspection

◆ Oil moisture > 1000 ppm

→ Recommend oil change

◆ Partial discharge > 100 pC

→ Stator inspection needed

◆ No maintenance for 3+ years

→ Planned maintenance overdue

CREW OPTIMIZER · MULTI-FACTOR

Certification

Availability

Distance

Response History

RANKED OUTPUT — TR034

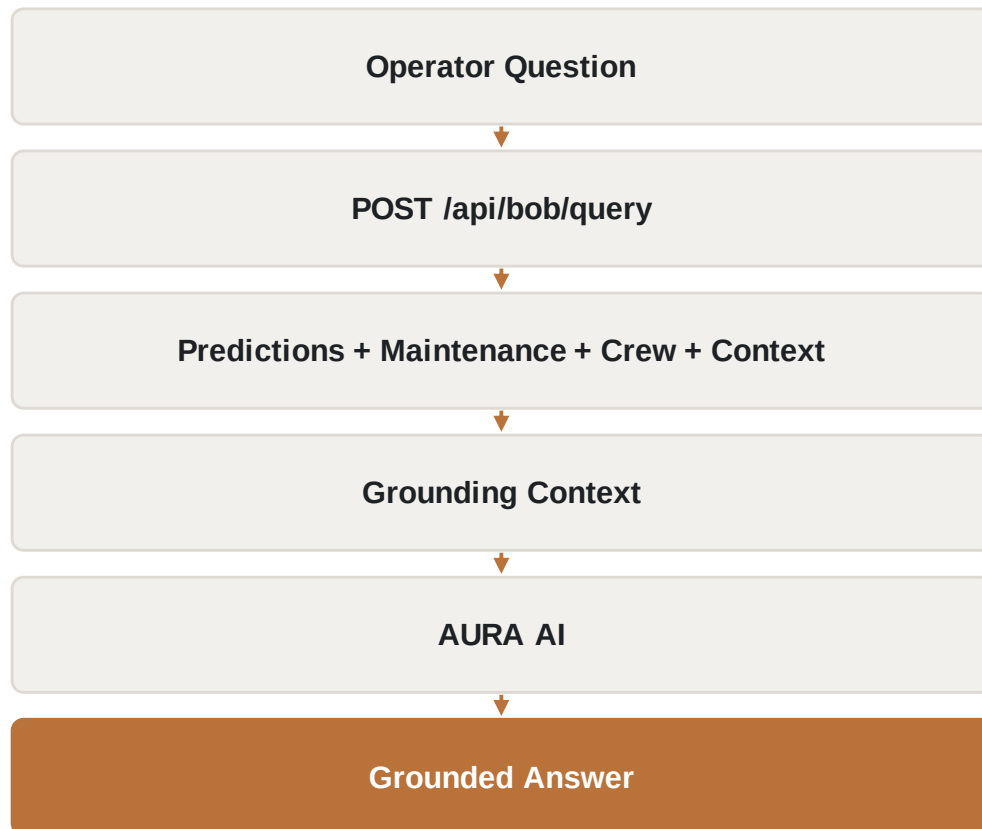
Crew Alpha-2 92%

Crew Bravo-5 81%

Crew Delta-1 68%

Grounded answers, not generative guesswork

IBM Bob is the AI-assistance layer operators talk to. Every answer is grounded in the project's current predictions, maintenance recommendations and crew status — never invented.



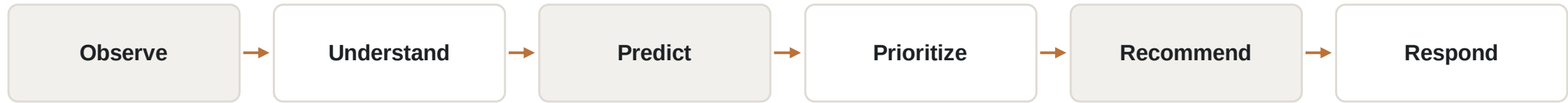
ASK AURA — LIVE EXAMPLE

Why is TR034 risky?

TR034 is HIGH risk (0.72). Drivers: rising oil temperature over the last 48h and vibration above baseline. Maintenance advisor recommends a thermal + mechanical inspection. Crew Alpha-2 is the best available match.

If the data isn't there, AURA says so — it never invents an asset, a reading, or a crew.

Beyond the hackathon



learns from outcomes — feeds back into Observe

MEASURED AGAINST



WHAT WE'RE HONEST ABOUT

- ◆ The model needs validation on real utility data before any production claim.
- ◆ The heuristic fallback is a resilience mechanism, not a substitute for a validated model.
- ◆ Crew and weather recommendations are only as good as the roster and forecast data supplied.



THANK YOU

Let's keep the **grid** a step ahead.

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